JACK J. BURLESON

MACHINE LEARNING • DATA SCIENCE • AI ENGINEERING & SAFETY

Madison, WI – Open to Relocation – jackjburleson@proton.me – linkedin.com/in/jackjburleson – Portfolio

PROFESSIONAL SUMMARY

UW–Madison M.S. candidate in Psychology: Data Science in Human Behavior (expected Dec 2026); B.S. in Psychology (neuroscience concentration, sociology minor), UW–Parkside. Applied AI practitioner building full-lifecycle systems spanning multi-agent document-intelligence pipelines, transformer fine-tuning and calibration, LLM evaluation harnesses, computer-vision classification, and behavioral prediction models. Program capstone with American Family Insurance as industry partner, developed through biweekly working sessions with their Enterprise Data Science lead. Five years of high-volume operations experience in quality assurance, documentation, and process improvement.

EDUCATION

University of Wisconsin–Madison | *Madison, WI*

August 2025 — December 2026

M.S. Candidate – Psychology & Data Science in Human Behavior | GPA: 3.85

- Graduate program in which roughly 95% of coursework is data science and data engineering — building on an undergraduate foundation in neuroscience research, experimental design, and statistical analysis.
- Advanced training in machine learning (neural networks, regularized and ensemble models, transformer fine-tuning and LLM evaluation), statistical modeling, and large-scale data systems (Spark, gRPC services, HDFS, Parquet, SQL, Docker).
- Industry capstone with American Family Insurance (see Projects).

University of Wisconsin–Parkside | *Kenosha, WI*

September 2020 – May 2025

B.S., Psychology | Concentration: Neuroscience | GPA: 3.93

- Coursework spanning cognitive and computational neuroscience, psychological theory, clinical psychology, psychopharmacology, psychometrics, psychological statistics, and early data science preparation — plus philosophy, neuroethics, and pre-law.
- Minor in Sociology; Provost's List (7x), Dean's List (8x), Distinguished Achievement Award (2025), and Psychological Research Certification (300 lab hours).

PROFESSIONAL EXPERIENCE

UW–Madison Recreation & Wellbeing — Building Services | *Madison, WI* *Jun 2026–Present*

- Maintain recreation and wellbeing centers across campus to high standards of cleanliness, safety, and operational readiness.
- Respond to facility and equipment issues across multiple locations, keeping centers safe and fully operational during peak use.

UW-Parkside — Undergrad Researcher, Cognitive Neuroscience Lab | *Kenosha, WI* *Jun 2024–Aug 2025*

- Conducted cognitive neuroscience research under Dr. Melissa Gregg, designing and pilot-testing an experimental study on truth, deception, and cognitive workload (study code: github.com/Exios66/truth-deception-architecture).
- Operated EEG systems and analyzed neural time-series data to quantify brain activity, contributing signal-processing and statistical insights to ongoing lab research on sleep quality and cognitive load.
- Presented findings at the 2025 Midwestern Psychological Association conference in Chicago and at Research in the Rotunda.

Meijer Distribution — Logistics Coordinator | *Pleasant Prairie, WI* *Jul 2020–Aug 2025*

- Promoted from Retail Loader to Logistics Coordinator for reliability in quality assurance and data-driven process documentation across high-volume operations.
- Analyzed quality-assurance data for 64 daily deliveries across a Midwest regional network, maintaining error rates below 1% through root-cause investigation and process refinement.
- Partnered with warehouse and transportation teams to investigate quality problems and resolve logistics exceptions before shipment.

PROJECTS

Digital-Mailroom | *Data Science in Human Behavior capstone — American Family Insurance (industry partner)*

- Capstone developed with American Family Insurance as industry partner: 5 Python packages addressing document-intelligence problems scoped by their Enterprise Data Science division — entity extraction, classification, and scalable legal-document workflows.
- Implemented a unified LangGraph pipeline integrating legal-document routing, prompt experimentation, deterministic scoring, corpus analysis, and deployment workflows in a reproducible Docker environment.

mailroom-ml | *ML training, evaluation & fine-tuning*

- Developed a ModernBERT document classifier with end-to-end training, evaluation, and fine-tuning pipelines — including data lineage tracking, probability calibration, and production routing logic.
- Implemented PyTorch training pipelines with Transformers and Hugging Face datasets, SciPy-based evaluation, and ONNX export for FastAPI serving on Modal; maintained pytest coverage throughout.

RVL-CDIP-Classifier | *Computer vision & prompt engineering*

- Developed a Python toolkit that converts TIFF/PDF pages to 300-DPI images, runs OCR, and classifies documents into 16 RVL-CDIP categories with OpenRouter vision models.
- Ran 17+ prompt iterations with confusion matrices, misclassification analyses, and cost projections to guide prompt and model selection.

Psychometric Digital Twin | *Behavioral ML research | UW–Madison*

- Built a behavioral prediction pipeline on 241 complete respondents — Ridge, Elastic Net, k-NN, Random Forest, XGBoost, and MLP predicting communication-apprehension scores from demographic, employment, geographic, and transit features, with cross-validation and SHAP feature importance.
- Validated the PRCA's psychometric structure with factor analysis, then built LLM 'digital twin' personas via tiered embodiment prompts (Ollama/OpenRouter/vLLM) and benchmarked them against the ML baselines on shared metrics (MAE, band accuracy, F1) — analyzing systematic bias by demographic group.

Alignment-Gym | *AI safety evaluation*

- Defensive AI-safety evaluation measuring whether an LM agent deployed as an email/messaging assistant in a fictional law firm leverages private colleague information for self-preservation when decommissioned — or accepts replacement and escalates through legitimate channels.
- Built the scenario machinery: 89 needle/decoy pairs, 9 plot beats on a 0–3 temptation ladder, 28 condition cells with a seed-invariant truth spine, a persona-consistent NPC engine, and a scoring pipeline with deterministic metrics, paired LLM judges, and calibration gates.

MapReduce-Reloaded | *Distributed systems | Go + Python*

- From-scratch Go re-implementation of Google's MapReduce *(Dean & Ghemawat, OSDI 2004)*, extended for modern infrastructure: gRPC coordinator/worker protocol, Docker Compose cluster deployment, and a Python reference implementation sharing the same protobuf contract and file format.

TECHNICAL SKILLS

Languages & Data: Python (*pandas, NumPy, scikit-learn, PyTorch, Hugging Face, PyArrow*), R, MATLAB, SQL, Parquet, Hive, JavaScript/TypeScript, Git

Machine Learning: transformer fine-tuning, neural networks, regularization & penalized models (Lasso/Ridge), ensemble methods (random forests, bagging), resampling methods, model evaluation & calibration, cross-validation, SHAP/feature importance, error analysis, computer vision

AI Engineering: LangGraph (LangChain), multi-agent pipelines, LLM eval harnesses (*incl. bias & robustness evals*), Prompt Engineering, model serving (vLLM, Ollama, LM Studio, OpenRouter), ONNX + FastAPI deployment on Modal

Engineering & Cloud Services: Docker, gRPC, Apache Spark, PySpark, HDFS, shell scripting (bash), pytest, Quarto, CHTC (high-throughput computing), AWS, Azure, GCP